Synthesized Features

A persona framework for collections allocation

Group Retail Risk · Group AI, Data & Analytics · November 2026

SUMMARY

Synthesized features give allocation a view of the customer, built from data we already hold

TODAY

No model beyond 60 DPD. Cases are allocated by bucket and product, and late buckets hold little behaviour to model.

UC2 · SYNTHESIZED FEATURES

Signals from every data domain, condensed each month into four answers per account:

1. **How did they get here?**
2. **Can we reach them?**
3. **Will they pay?**
4. **Can they pay?**

WHAT IT SHOWS

54% of card collections came from just a fifth of the accounts, picked out in advance by the framework.

8× Within the same DPD bucket (91–120), some loan customers pay 8 times as often as others.

12% of 180+ accounts show a positive signal; one in five of them pays, against under 3% of the rest.

NEXT

By 20 Nov
UC2 synthesized features signed off; one may follow later

20 Nov – 11 Dec
UC1 transcript signals and UC2 combined, with action support

11 – 20 Dec
Pilot selection

From 4 Jan
Pilot

WHERE WE START

Allocation follows the bucket, not the customer

WHO WORKS THE CASE

In-house or agency

Rule-based: **DPD level and product** decide which team gets the case, not whether the customer can pay or has kept their word.

WHAT WE OFFER

Settlement, plan or push back

Pre-approved settlement programmes (PLUS, CARES) with waivers fixed **by product and DPD stage**, not by the customer's capacity or track record.

WHERE 180+ EFFORT GOES

374k accounts at 180+

Worked as one pool, though a small share still shows signs they could pay. Finding them is the 180+ goal.

SYNTHESIZED FEATURES

Everything we know about a customer, condensed into a few states

DATA DOMAINS

- Card billing
- Card transactions
- CASA transactions
- Salary
- Loan instalments
- Bureau
- Digital logins
- Collector activity
- Customer records
- Transcripts · joining

→

SIGNALS

- *Is salary regular?*
- *Do calls connect?*
- *Are promises kept?*
- *Last seen abroad?*
- *Paying while late?*

and 45+ more

→

PERSONA STATES

A1 · Deterioration
How did they get here?

A2 · Locatability
Can we talk to them?

A3 · Payment credibility
Will they pay?

A4 · Ability
Can they pay?

UC1 conversations · joining

10 data domains · 50+ signals · 4 states per account, every month

61–180 DPD: personas that drive allocation

180+ DPD: find the customers with a positive signal

THE IDEA

Three questions before a case reaches a collector

1 How did they get here?

A1 · Deterioration

- **Curing now** — stabilized
- **Cured or paid before** — slow roller · repeat delinquent
- **Paid, then stopped suddenly** — seasoned · early-life straight roller
- **Never paid** — early payment default

2 Can we talk to them?

A2 · Locatability

Two checks
where they were last seen · whether calls connect

- **Reachable** — calls connect
- **Non-Reachable** · calls fail
- **Skip** · last seen abroad

3 What can we offer?

The nuance, for every customer

A3 · Credibility · Will they pay?
High · Moderate · Low · Unknown

A4 · Ability · Can they pay? IN PROGRESS
High · Moderate · Low · Unknown

UC1 · Conversations · joining
intent and ability to pay, from call transcripts

INFORMS
Context and priority

INFORMS
How we can reach them

DECIDES
Offer and agent

HOW IT IS BUILT

Built to be explained and trusted

Rules you can trace

Every state comes from a written rule, and the record says which rule assigned it and why.

Unknown is a state

When evidence is missing we say so, with the reason, instead of guessing.

Point in time

Each month end uses only what was known that day: the view a strategy would have had.

Customer-level evidence

Salary, CASA, bounces and logins are read across all of a customer's accounts, not only the delinquent one.

PART 1

The four axes

Deterioration · Locatability · Payment credibility · Ability

A1 · DETERIORATION STATES · DECIDED BY RULES

Customers curing now pay far more often than any other group

HOW A CUSTOMER GETS A LABEL

Read from the DPD history alone, so it covers every account. Asked in order; the first "yes" decides.

1	Nothing paid since the first months?	→ **Early payment default** bad from the start
2	Holding at one level for months?	→ **Stabilized** stopped falling
3	Back within a year of curing?	→ **Repeat delinquent** been here before
4	Fell to 61+ fast, after a year of good history?	→ **Seasoned straight roller** sudden stop
5	Fell to 61+ fast, in the first year?	→ **Early-life straight roller** fell fast, early
6	Otherwise	→ **Slow roller** fell slowly, paying on the way

AUGUST MIX · SHARE OF ACCOUNTS



60% of card accounts curing now paid within 3 months; of those who only cured before, 22% did.

Half of loans (six in ten cards) paid for a while, then stopped all at once: an event, not a habit.

31 August 2026, 61–180 DPD · paid: meaningful payment within 3 months

A2 · LOCATABILITY STATES · DECIDED BY RULES

Only one in five card customers at 61–180 can be reached

HOW A CUSTOMER GETS A LABEL

Location Where were they last seen?

Logins and in-person card use show where they are; a travel purchase or collector note signals they have left. The latest evidence wins.

In Country | Skip | Unknown

Reachability Do our calls connect?

Recent call days all failed, no contact since → Non-Reachable. A recent contact → Reachable.

Reachable | Non-Reachable | Unknown

Informs how we work a case: by phone, or with tracing.

AUGUST MIX

Location × reachability, 61–180

% of card accounts · loans below

	Reachable	Non-Reachable	Unknown
In Country	8.5% loans 10.2%	13.6% loans 14.2%	1.9% loans 2.7%
Skip	6.2% loans 8.1%	31.7% loans 28.5%	6.1% loans 6.8%
Unknown	6.3% loans 7.9%	22.3% loans 14.9%	3.4% loans 6.8%

1 in 3 look abroad and do not answer, the largest group on cards and loans alike: candidates for tracing or an agency.

3.7% of 180+ card accounts are reachable (loans 4.8%), a positive signal for finding those still worth working.

A3 · PAYMENT CREDIBILITY SCORE · POINTS → BAND

Within the same bucket, credible customers roll far less

HOW A CUSTOMER GETS A LABEL

Three sources earn points; we add them up

Promises kept
last 12 months
Kept +2 | Partly +1 | Broken −1

Paying while delinquent
regular = the monthly due paid in 2 of the last 3 months
Regular +2 | Some +1

Record before
clean = paid as billed before falling behind
Clean +1

High	Moderate	Low
4+ points	2–3 points	0–1 points

> An explicit refusal to pay → **one band down**

AUGUST MIX

	High	Moderate	Low
Cards 61–180	2.5%	19.4%	78.0%
Loans 61–180	4.9%	27.9%	67.2%
Cards 180+	0.1%	2.7%	97.2%
Loans 180+	0.3%	4.3%	95.4%

42% vs 90% of card customers at 91–120 DPD rolled further: High vs Low credibility.

59% of card customers made no promise to pay in the last 12 months: their credibility rests on payments alone.

Roll rates: back-test, 3-month outcomes compared within each DPD band

A4 · ABILITY IN PROGRESS

Ability tells us what a customer can realistically be offered

HOW A CUSTOMER GETS A LABEL

Four evidence families; we count how many are strong*

Income
Regular salary, ours or bureau

Money in
Money arriving or held, not salary

Spending
The customer's own spending

Other products
Activity on their other products

High	Moderate	Low	Unknown
2+ strong	1 strong	some evidence, none strong	no evidence

A cheque bounce or returned direct debit → **one band down**

AUGUST MIX

	High	Moderate	Low	Unknown
Cards 61–180	1.6%	13.9%	27.3%	57.2%
Loans 61–180	1.2%	8.8%	49.3%	40.7%
Cards 180+	0.3%	6.0%	8.6%	85.1%
Loans 180+	0.1%	3.5%	11.0%	85.4%

Needs: GRR's bureau salary feed

About a third of customers have no CASA with us, and half never had a salary credit here. Without a regular bureau salary feed they stay Unknown; with it, income becomes visible across the book. Today the bureau's bulk salary pull reaches about 1% of customers.

57% of card customers at 61–180 show no evidence of their money yet.

4× High-ability card customers are back below 61 DPD three months later four times as often as those with no evidence.

* Strong = regular and above the floor in at least 2 of the last 3 months (floor: AED 3,000 or twice the amount due); at 180+: 3 of the last 6 months

PART 2

From states to personas

Too many personas · where the value sits · what we hand over

COMBINING THE STATES

Combined, the states give hundreds of personas: far too many to give each a strategy

6 deterioration states
A1

×

9 locatability cells
A2

×

3 credibility bands
A3

×

4 ability bands
A4

=

648 possible personas
today

×

UC1 transcript signals
intent · hardship · dispute

393
occur among card accounts in August

222
have enough accounts (10+) to judge

A **persona** is one combination of states. Each UC1 signal multiplies the count again, and many personas hold only a handful of accounts.

Cards 61–180 · August 2026 · credibility High / Moderate / Low; ability includes Unknown

WHERE THE VALUE SITS

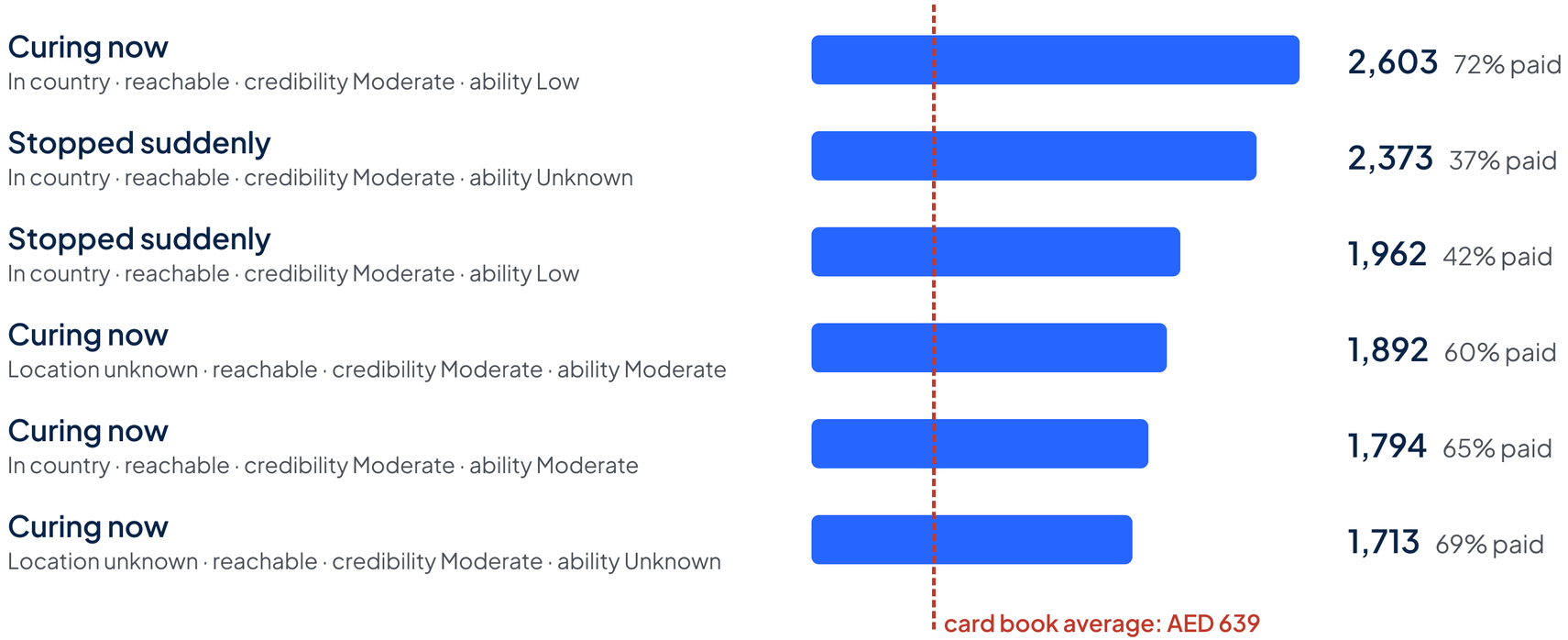
A few dozen personas bring in over half the money

54%

of the money collected came from **one fifth of card accounts**, picked in advance by their persona

That fifth sits in a few dozen personas, mostly customers **curing now or cured before** who show **some credibility or ability**. The examples collect about three to four times the book average.

EXAMPLES · AED COLLECTED PER ACCOUNT IN 3 MONTHS



Cards 61–180 · AED per account: what the persona's May 2026 accounts paid in June–August, per account · 54%: tested on months and accounts not used to rank them

THE DELIVERABLE

What we hand over: every account's persona, every month

1

Persona table

Every account, every month: its four states, each with the reason, and its persona

2

Persona profiles

For every persona: how many accounts, how often they paid and what they collected, back-tested on past months

3

Model specifications

One document per feature, with the full logic and back-test results

4

UC1 signals

Joining from December: what customers tell us on calls, on the same accounts

GRR DECIDES which personas get which offer, agent and channel

Optional: how we would use the data to narrow the strategy question →

PART 3 · OPTIONAL

How we would approach allocation

Only in-house needs a strategy · tiers from what personas paid · keeping them stable

THE SIMPLIFICATION

Only the cases we keep in-house need a strategy of their own

EVERY CASE

61–180 DPD

allocated first, then worked

→

In-house — STRATEGY PER PERSONA

Offer, agent and channel chosen for who the customer is

Agency — POLICY TERMS

Worked by the agency, paid on cash collected

Non-workable pool — POLICY TERMS

Least probable to pay, referred out under the policy

THE QUESTION CHANGES

FROM

~~A strategy for every persona~~

TO

Which personas earn in-house effort, and what we do with them

Agency and non-workable handling as in the collections policy (FR Policy on Operations, July 2025)

A FIRST LOOK

How an account gets a tier: by what its persona actually paid

1 Describe each account

One state on each of the four axes. Together they form its **persona**.

- **Stopped suddenly**
- **Skip · Non-Reachable**
- **Low credibility**
- **Unknown ability**

2 Measure what it paid

For each persona: every account in May 2026, everything it paid in June–August, averaged per account.

Example: 400 accounts paid AED 70,000 in total → **AED 175 per account**. 10% of them paid, AED 1,750 each on average.

Observed, not modelled: the average combines how often and how much.

3 Rank and cut

The personas with at least 10 accounts (97% of cards), ranked by AED per account:

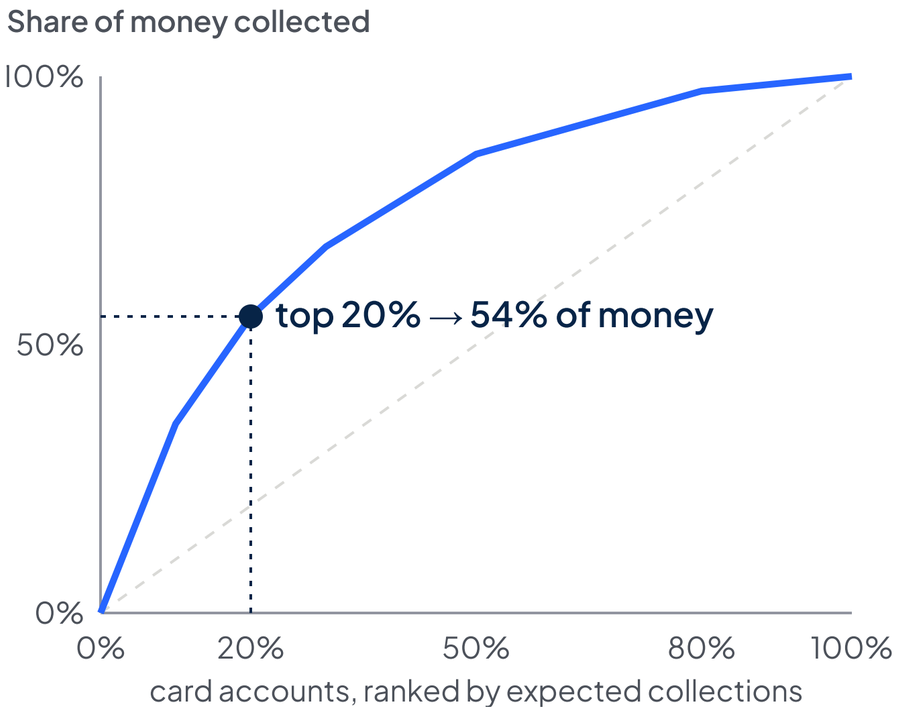
In-house	AED 1,250 and above
Agency	between
Non-workable	under AED 175

Illustrative cut-offs from the data; GRR would set them from the commission rate card and in-house cost.

Cards 61–180 · back-test May → Jun–Aug 2026; tiers hold on unseen months · personas under 10 accounts unassigned (3%)

A FIRST LOOK

In-house would take a sixth of card accounts and half the money collected



Tier	Accounts	Money collected	Recovery, 3 m	Call days
In-house	17%	48%	10.0%	11.7
Agency	54%	46%	2.7%	11.8
Non-workable	29%	6%	0.56%	10.9

In-house: mostly customers curing now or cured before, who are credible or able.

Non-workable: mostly stopped suddenly or never paid, and not reachable.

10.9 call days per account in 3 months still go to the non-workable pool, almost as many as in-house.

Cards 61–180 · table out-of-sample (5 folds, Jan–May 2026); call days: August · cuts AED 1,250 / 175 · loans: same order, thresholds less stable

WORK IN PROGRESS

Keeping the tier list stable from month to month

WHY TIERS MOVE

35%

of loan accounts change tier from one month to the next while their persona stays the same (cards 13%)

Most of it is chance: many personas hold only a handful of accounts, so one large payment moves their average across the cut. Two random resamples of the same month already disagree on 27% of loan tiers.

WHAT WE TRIED · TIER CHANGES, PERSONA UNCHANGED

	CARDS	LOANS
Today: one month per persona	13%	35%
Several months pooled	6%	15%
Pooled, with a band around the cut	5%	14%
Pooled, large payments capped	7%	11%
Pooled, on broader groups	1%	5%
A moving or rank-based cut alone	12–13%	25–34%

None of them lost money collected at the same in-house share.

NEXT

Longer pooling
12 months, with history from before 2026

Broader groups that keep location
fewer, larger groups for the tier; the full persona for the strategy

Re-run every 3 months
with triggers for an early re-run: in-house paid rate or the book's collection level leaving its usual range

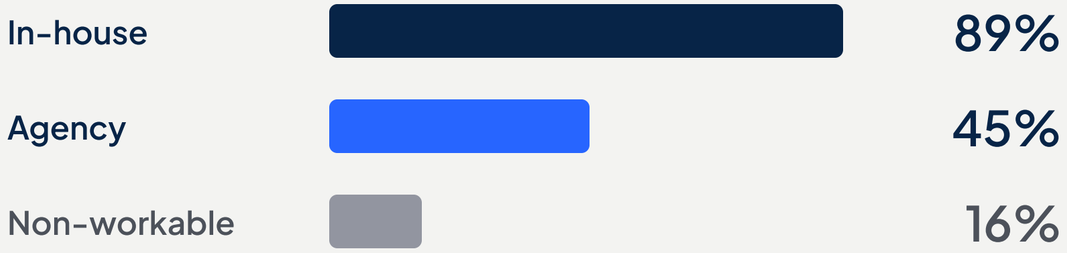
Confirm on later months
once payment data runs past August

Cards and loans 61–180 · cohorts January–May 2026 · tier built on earlier months, read on the next · broader groups: family × reachability × credible or able

UC1

In-house is where we talk to customers, so that is where UC1 adds its signals

RIGHT-PARTY CONTACT IN THE LAST 3 MONTHS · CARDS



All card accounts 41% · loans 52%

WHAT TRANSCRIPTS ADD

- Stated intent to pay
- Change in income or job
- Hardship reason
- Dispute
- Promise made on the call

> These come from what customers say on calls, so most of them will come from the in-house cases.

Cards 61–180 · August 2026 · groups at the illustrative cut-offs

TOWARDS THE STRATEGY WORKSHOP

The in-house personas become a short list for GRR to give each a strategy

THE FIVE LARGEST IN-HOUSE PERSONAS · CARDS

PATH	WHERE · REACH	CREDIBILITY	ABILITY	SHARE
Stopped suddenly	In country · **reachable**	Low	Unknown	**8.0%**
Cured before	In country · **reachable**	Low	Low	**6.4%**
Curing now	Unknown · **reachable**	Moderate	Unknown	**5.0%**
Stopped suddenly	In country · **reachable**	Moderate	Low	**4.9%**
Cured before	In country · **reachable**	Moderate	Low	**4.7%**

26 of 51 in-house personas cover 80% of in-house accounts

1 **The data proposes**

In-house personas, ranked by size and what they pay. All reachable; credibility and ability vary

2 **UC1 refines**

Transcript signals on top, where we talk to customers

3 **GRR decides**

Offer, agent and channel for each persona on the list, in the strategy workshop

Cards 61–180 · August 2026 · share of in-house accounts · in-house at the illustrative cut-off of AED 1,250 per account in 3 months · loans: 11 of 16 cover 80%

APPENDIX

Signals, rules and governance

For the detailed discussion: where each signal comes from, the exact rules, and how the framework is controlled.

APPENDIX · A FIRST LOOK

What each group could mean, for discussion

In-house

Curing now or cured before, reachable, credible or able

Convert through skilled collectors: settlement, restructure or deferral

Policy tools: CARES settlements, loan restructure, deferral

11% of cards · AED 40m

50% paid in 3 months

Agency

Stopped suddenly or cured before, not reachable

Place with agencies at 61–180, skip tracing where they look abroad

Policy: referral by product, DPD stage, contactability and strategy

54% of cards · AED 213m

17% paid in 3 months

Non-workable

Stopped suddenly or never paid, not reachable

Free collector time: refer to agencies or stop calling

Policy: non-workable pool to agencies (HOD approval), or no calling with dunning as for Dead Beat accounts

32% of cards · AED 138m

4% paid in 3 months

180+ The 12% with a positive signal: lift out of the pool for an in-house attempt. About one in five pays, against under 3% of the rest.

For discussion in the strategy workshop; not agreed with GRR · cards 61–180; AED = outstanding principal; paid = meaningful payment

APPENDIX · A FIRST LOOK

How a pilot could test it, if the strategy workshop takes it forward

Test in each tier

- **In-house:** skilled collectors and settlements vs business as usual
- **Agency:** placement at 61–180 vs in-house calling
- **Non-workable:** agency referral or dunning only vs business as usual (hold-out)
- Random split within each tier, 3 months

Measure

- Meaningful payment and roll rate
- Amount collected per account
- Recovery of outstanding principal
- Call days and cost per AED collected
- 180+: the flagged group vs the rest

Size and guardrails

- Cards: agency and non-workable fill both arms in one month
- In-house needs a 10–pt target; loans need 10 pts and pooling
- Same policy and compliance rules
- Monthly readout and a stop rule

Sample per arm (cards, 5–pt lift): agency 970, non-workable 375; in-house 388 at a 10–pt lift · two-sided α 0.05, 80% power

APPENDIX · RISK SCORE

Personas rank payers far better than the bucket; a model adds a little on top

How well each view ranks who pays (AUC) · 61–180

Ranking customers by	Cards	Loans
DPD bucket only (today)	0.65	0.62
Personas	**0.83**	**0.77**
Personas × DPD bucket	0.84	0.77
Simple model (axis states + bucket)	0.84	0.78

Out-of-sample: 5 folds by account, January–May 2026 cohorts. 0.5 = chance, 1.0 = perfect.

Personas beat the bucket in every test

+0.17 on cards, +0.14 on loans. The top 20% of card accounts by persona hold 54% of the money collected (bucket: 30%).

The bucket adds almost nothing within personas

+0.01 on cards, +0.00 on loans.

A model on the same inputs adds about **+0.02** over the personas. Risk score modelling is planned for **Phase 2**.

Paid = meaningful payment within 3 months (the risk score's target) · strict and rolling time splits give the same picture

APPENDIX A

Signals and their sources

Axis	Signals	Sources
A1 Deterioration	Monthly DPD bucket path, restructures linked to their original loans	Way4 billing history, loan master
A2 Locatability	UAE vs foreign logins, in-person card use (POS, ATM) by country, departure notes, travel purchases, call outcomes, deceased flags	Digital login, Way4 card transactions, CASA card transactions, DCORE follow-ups, customer records
A3 Credibility	Promises and plans with their outcome, payments vs dues per cycle, record before delinquency, refusals, disputes	DCORE plans and follow-ups, card statements and payments, loan monthly dues and cash
A4 Ability	Salary per month (larger of ours and bureau), CASA balance and activity, external card payments, cheque / direct-debit returns, bureau bounces	Wealth-management salary and CASA features, AECB salary and bounce data, loan master (instalment)

Customer-level signals are read across every account of the customer (CIF). Windows end at the observation month end: 3 months at 61–180 DPD, 6 at 180+.

All sources are existing bank datasets; no new data feed is needed to run the framework

APPENDIX B

The scoring rules at a glance

Axis	Evidence and points	State
A1 Deterioration	DPD history only, from January 2024 (reliable window)	First “yes” of six ordered questions (Appendix C)
A2 Locatability	Latest location evidence: foreign → Skip; UAE within 3 months (6 at 180+) → In Country; travel rule. Last 3 call days failed (6 at 180+) → Non-Reachable	Location × reachability (9 cells) + Deceased override
A3 Credibility	Promises: Kept ≥ 80% +2, Partly +1, Broken < 40% or last 3 broken –1. Payments: monthly due paid in 2 of the last 3 months +2, any payment +1. Record before: 80% paid as billed +1	High ≥ 4 · Moderate ≥ 2 · else Low; refusal –1 rung; no delinquency history → Unknown
A4 Ability	Four evidence families: income (salary, ours or bureau), money arriving or held, own spending, other products. In progress	High: 2+ strong · Moderate: 1 strong · Low: evidence, none strong; distress –1 band; no evidence → Unknown

Full specification and reference workbook per axis

APPENDIX C

How a journey state is assigned

#	Question	Yes when	State
0	Start of journey seen?	Oldest month already delinquent and not stabilized	Unknown
1	Early payment default?	≤ 3 months of history, rolled in, never paid since	EPD
2	Stabilized?	Pays about one month's due in 5 of the last 6 months (61–180 only)	Stabilized
3	Repeat delinquent?	Cured an earlier spell that reached 31–60 DPD, back within 12 months	Repeat delinquent
4	Straight roller?	Current to 61+ DPD in ≤ 3 months, no payment on the way	Seasoned (12+ months on book) or early-life
5	Otherwise	Paid something on the way down, kept rolling	Slow roller

Billing artefacts repaired first and counted · retail loans on the same tree, buckets shifted by one

APPENDIX · ABILITY

New ability rule vs the old income + CASA rule

SHARE OF ACCOUNTS IN EACH STATE, AUGUST

	High	Moderate	Low	Unknown
Cards 61–180	1.62% was 0.23%	13.9% was 14.4%	27.3% was 28.9%	57.2% was 56.6%
Loans 61–180	1.18% was 0.28%	8.8% was 8.9%	49.3% was 44.7%	40.7% was 46.1%
Cards 180+	0.27% was 0.03%	6.0% was 6.2%	8.6% was 6.4%	85.1% was 87.4%
Loans 180+	0.12% was 0.03%	3.5% was 3.5%	11.0% was 6.9%	85.4% was 89.5%

NEXT 3 MONTHS, 61–180*

	Cards			Loans		
State	Cure	Pay	Roll	Cure	Pay	Roll
High	31%	59%	53%	28%	75%	59%
Moderate	16%	41%	73%	15%	65%	69%
Low	11%	29%	82%	10%	45%	82%
Unknown	8%	20%	88%	9%	36%	85%

High is reachable: 2 of 4 strong families, not a CASA balance the bank sweeps.

Other products with us give evidence for customers with no CASA activity.

Only proven transactions count; bounces and reversals cancel out.

* Cure: below 61 DPD at month 3 · Pay: any payment, even partial · Roll: moves to a worse bucket

APPENDIX · CREDIBILITY

A 30-day window to keep a promise ranks customers best

DEADLINE FOR A KEPT PROMISE · RANKING POWER (AUC)*

	Cards 61–180	Loans 61–180	Cards 180+	Loans 180+
Next statement · loans: end of next month Today's rule	0.692	0.666	0.670	0.746
End of the month of the promise Proposed for loans and all 180+	same rule as today	0.643	0.665	0.736
30 days after the promise Lenient	0.714	0.666	0.690	0.746

Late payers are payers. Loan promises paid in the month after: 58% of these accounts then pay a monthly due in the next 3 months, the same as promises paid on time (58%); unpaid: 46%.

Customers moved down pay like the state they left. Loans moved from Moderate to Low by the month end: 49% pay a due, against 53% that stayed Moderate and 24% that stayed Low.

No record is the weakest part of Low. Accounts with no promise or payment record, mostly under 6 months on book, pay a due 4% of the time, against 12% for the rest of Low (cards 61–180).

* How well the credibility state ranks customers on paying a due in the next 3 months (0.5 = chance) · March–May 2026 · gaps ≥ 0.01 hold every month

APPENDIX D

Status and sign-off

Axis	Status
A1 Deterioration	**Closed** · signed off by GRR
A2 Locatability	**Closed** · signed off by GRR
A3 Credibility	Sign-off by 20 Nov
A4 Ability	Sign-off by 20 Nov, or later
UC1 transcripts	Joining, 20 Nov – 11 Dec

Status as of 11 Nov 2026